南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

生物医学工程系
Department of Biomedical Engineering

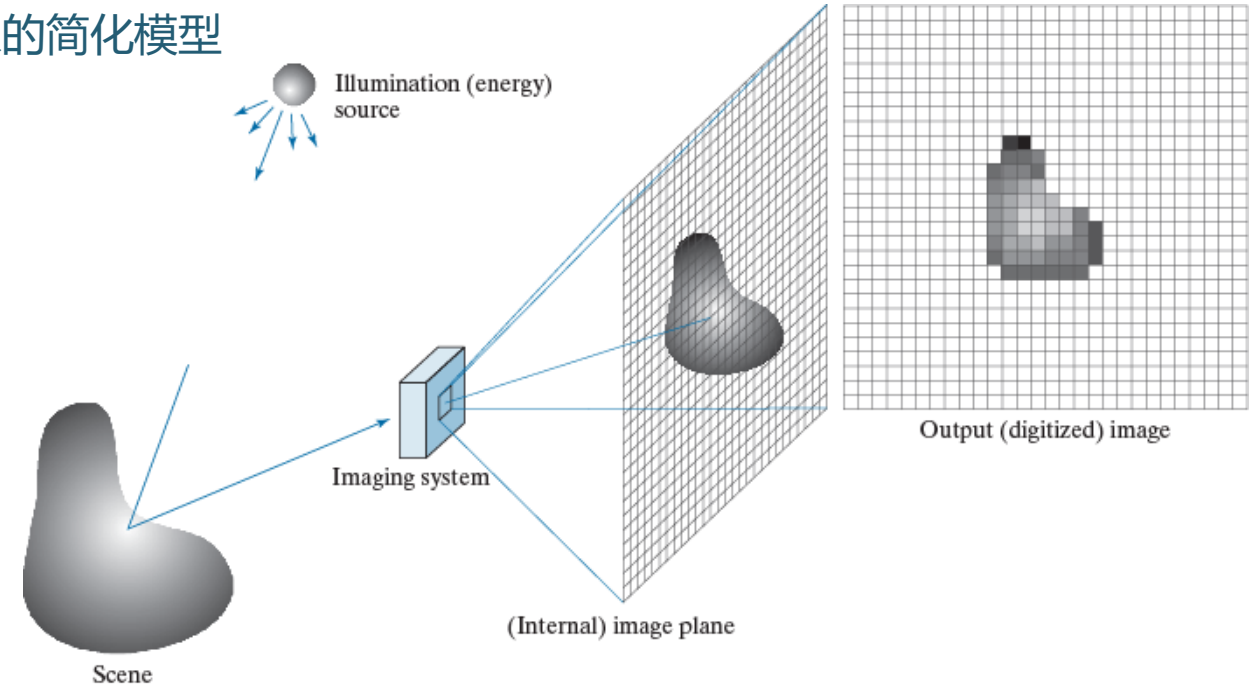
Digital Image Fundamentals

Yiming Li (李依明)
liym2019@sustech.edu.cn

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Imaging Formation Model

反射式光学成像的简化模型



$$f(x,y) = i(x,y)r(x,y)$$

Illumination Source: $0 < i(x,y) < \infty$

Image Objects: $0 < r(x,y) < 1$

Learning Objectives (Section 1)

Sampling: explain pixel spacing, spatial resolution, and aliasing.

Quantization: calculate gray levels and storage from bit depth.

Interpolation: select appropriate resampling methods for grayscale images.

Guiding question: do more pixels and smoother images necessarily make medical images more accurate?

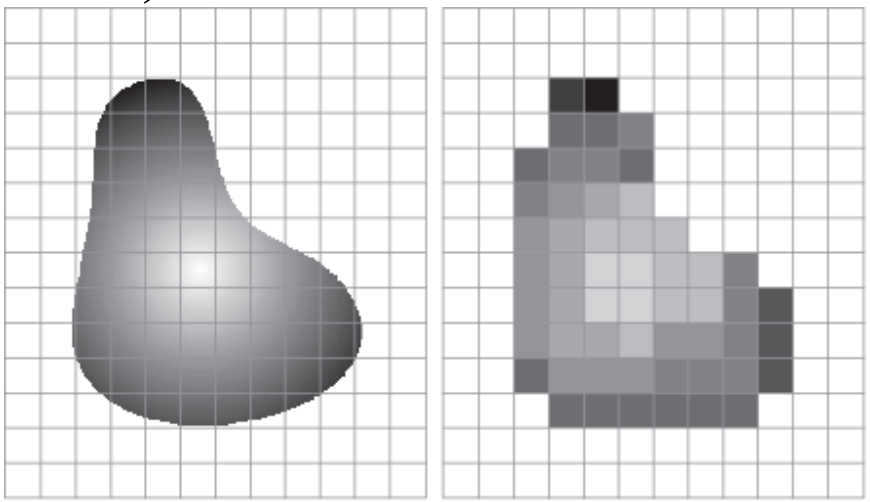
Preprocess of image formation

- **Sampling**

evaluate light intensity on a regular grid of points

- **Quantization**

map continuous signals to discrete values (natural numbers)



Sampling

- 2D grids used for sampling



rectangular hexagonal

- electronic cameras: rectangular, equidistant grids
- biology: hexagonal grids with varying resolution

Pixel Spacing and Spatial Resolution

Matrix size: the number of pixel rows and columns.

Field of view (FOV): the physical extent of the image.

Pixel spacing = FOV / pixel count along the same axis

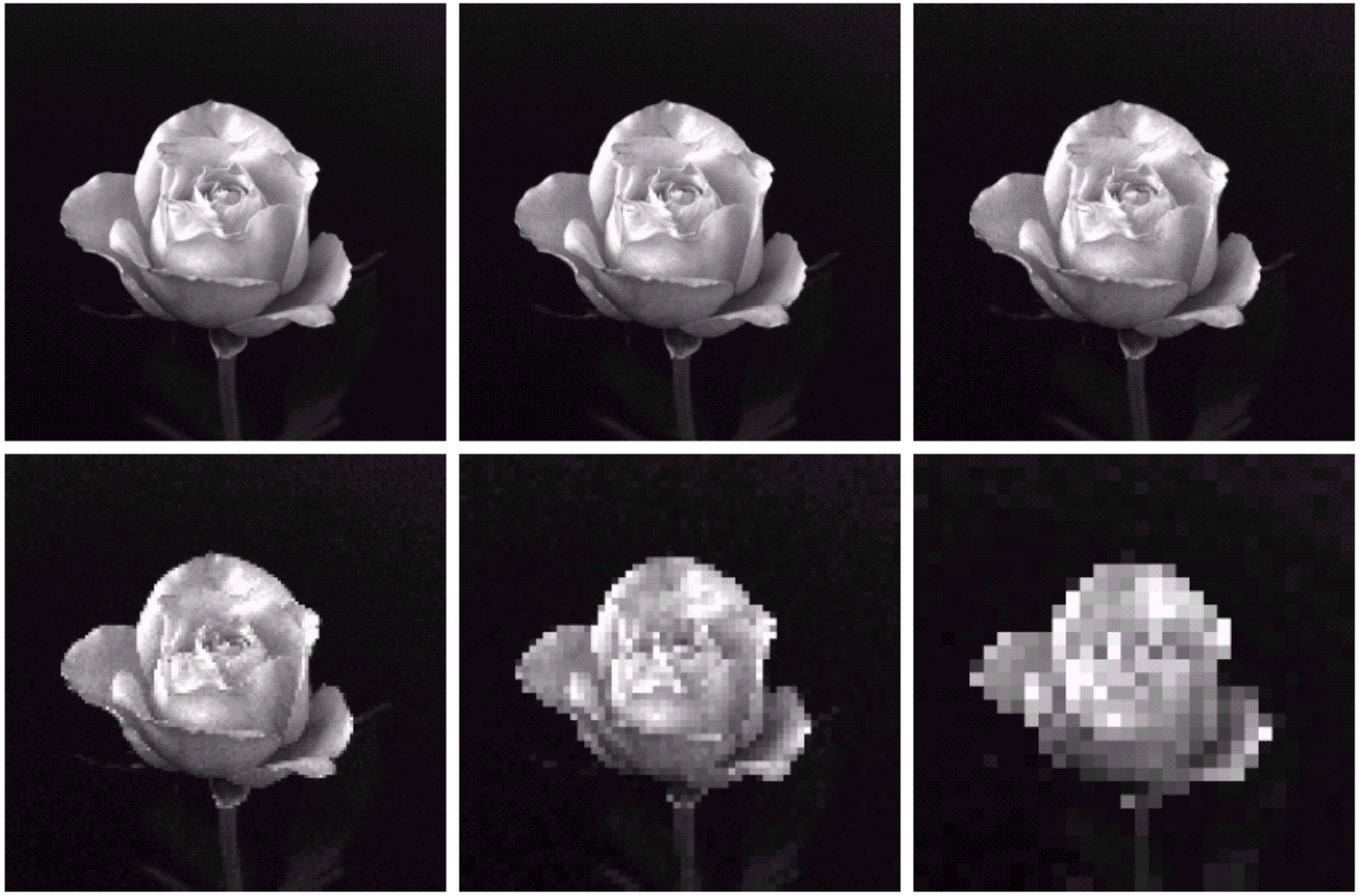
Example: 256 mm / 512 = 0.5 mm/pixel

Spatial resolution: the smallest distinguishable detail.
It depends on the imaging system, sampling, and reconstruction.

Does interpolation to 1024 × 1024 double the true detail?

Influence of Spatial Resolution

(a)原始图像(256×256)　(b)采样图像(128×128)　(c)采样图像(64×64)



(d)采样图像(32×32)　(e)采样图像(16×16)　(c)采样图像(8×8)

Sampling: Moiré Patterns

- Moiré patterns
  - sampling might cause artifacts

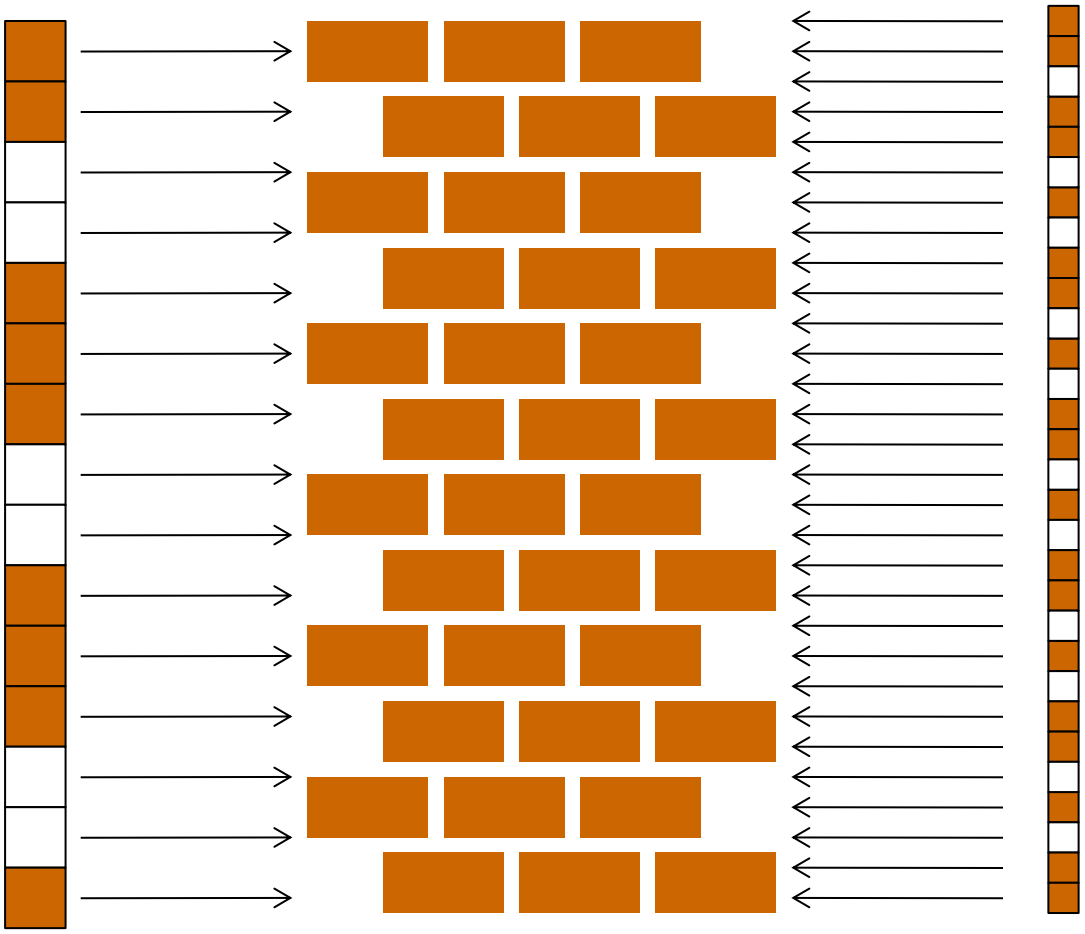


original picture



picture with Moiré pattern

Sampling: Moiré Patterns



sparse sampling
→ 4 layers

wall of bricks, 12 layers

dense sampling
→ 12 layers



Nyquist-Shannon Sampling Theorem

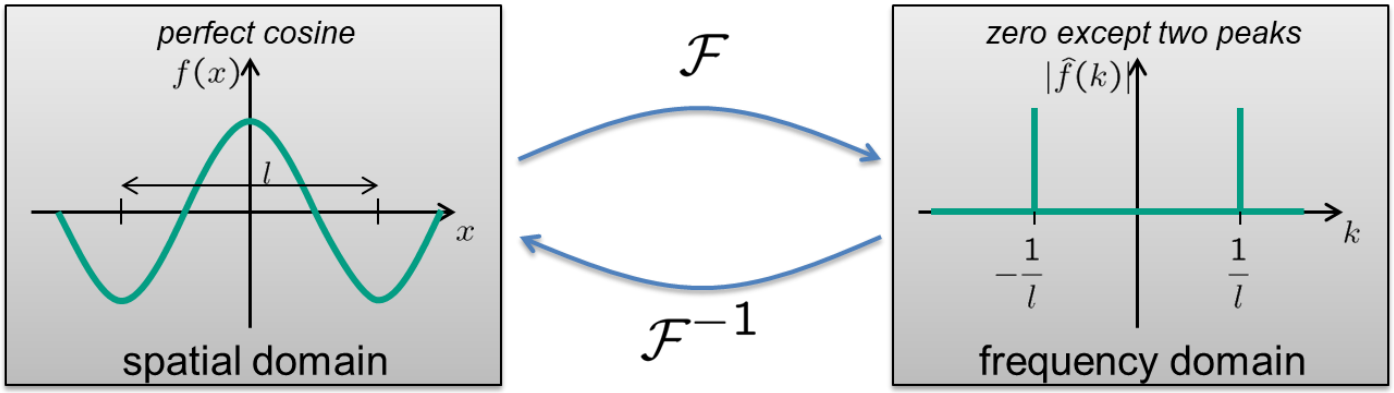
If f is band bounded signal with cutoff frequency k_0 then it is completely determined by giving its ordinates at a series of points spaced at most $\frac{1}{2k_0}$,i.e. the sampling frequency must be larger than $2k_0$

- Questions:
  - what is a band-limited signal?
  - what is a cutoff frequency?

Fourier Transform

空间域：信号随位置怎样变化?

频率域：包含哪些频率，以及各自的幅度和相位?



变化越快，对应的空间频率越高。

带限：超过最高频率 f_{max} 后，频谱为零。

采样需要足够密集，才能区分信号中的最高频率。

Nyquist–Shannon Sampling

采样率需要高于最高信号频率的两倍

$$f_s > 2\, f_{max}$$

等价地：$\Delta x < 1 / (2\, f_{max})$

前提：信号带限、均匀采样，并采用理想重建。
实际系统需考虑抗混叠滤波与采样余量。

f_s：采样频率　f_{max}：最高空间频率　Δx：采样间距

Questions

- Why can a low-resolution image on a smartphone still “look good” to the human eye, while a medical X-ray requires very high resolution to be useful?

- If you sample an image at too high resolution, what problems (besides larger storage) might you face?

Quantization and Bit Depth

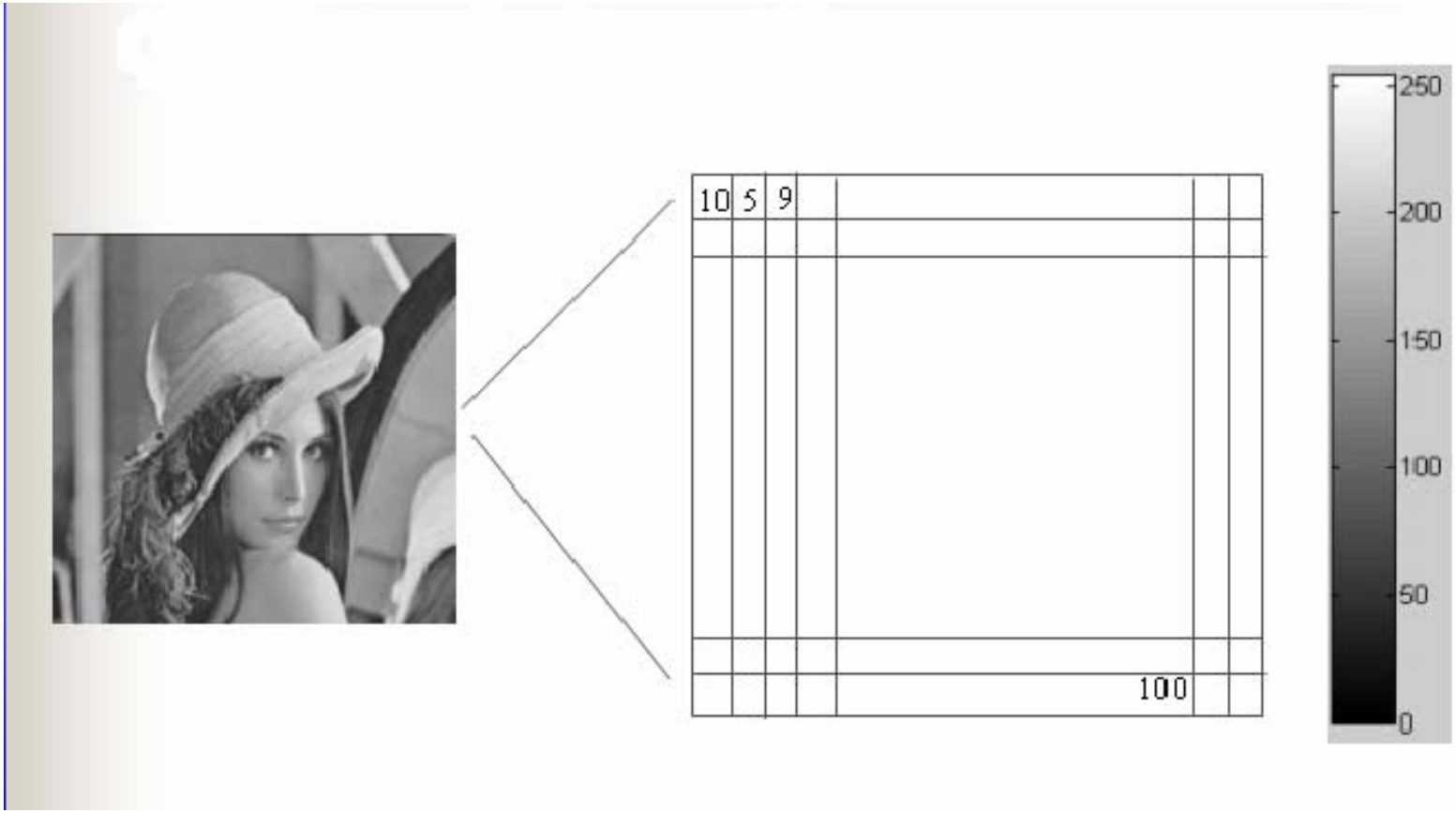
Quantization maps continuous amplitudes to discrete codes.

A bit depth of b gives $L = 2^b$ possible codes.

1 bit: 2 levels, usually coded as 0 and 1.
8 bits: 256 levels, usually coded as 0–255.
12 bits: 4096 levels, usually coded as 0–4095.

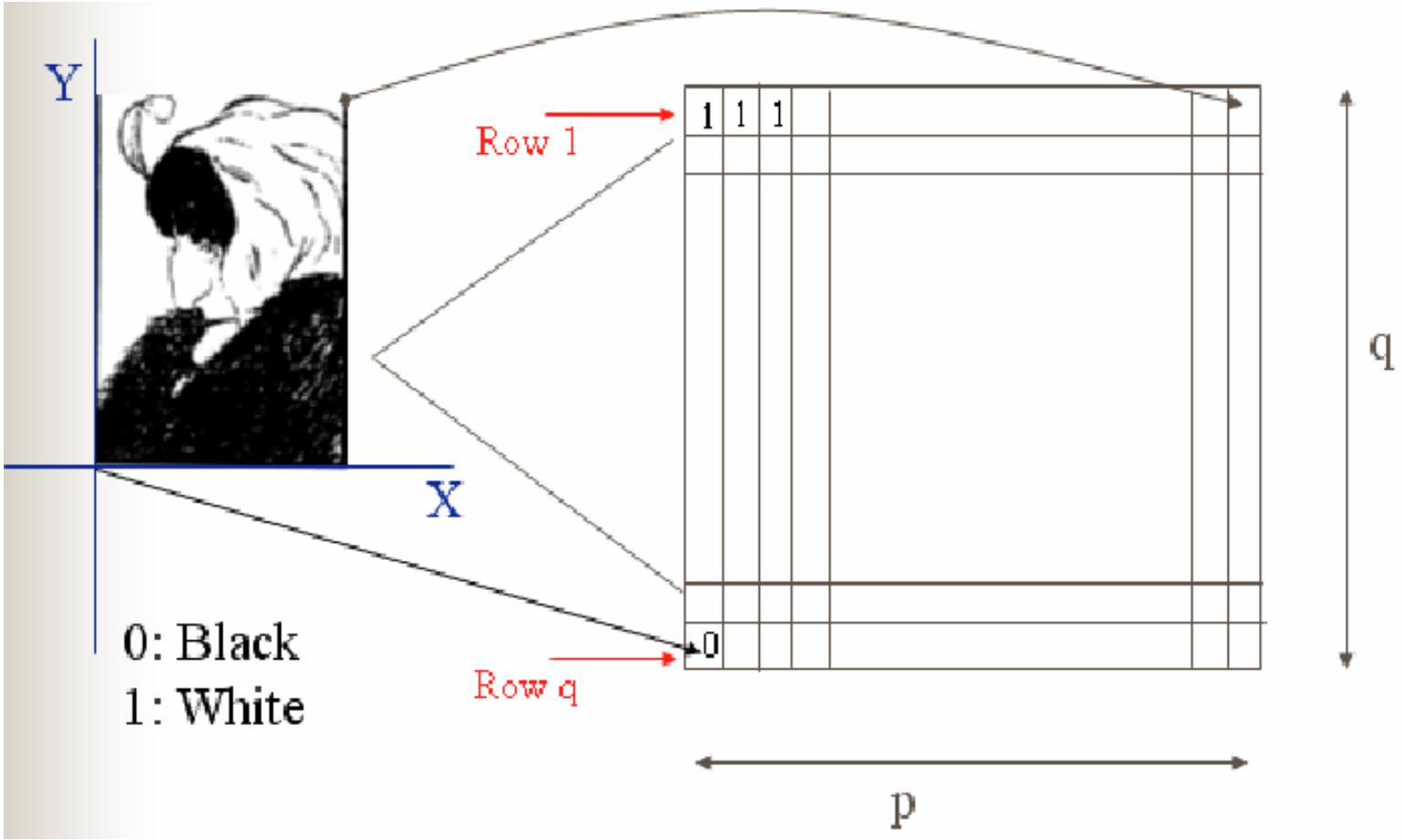
Bit depth sets the number of codes. Sensor dynamic range also depends on the saturation limit and noise floor.

Quantization

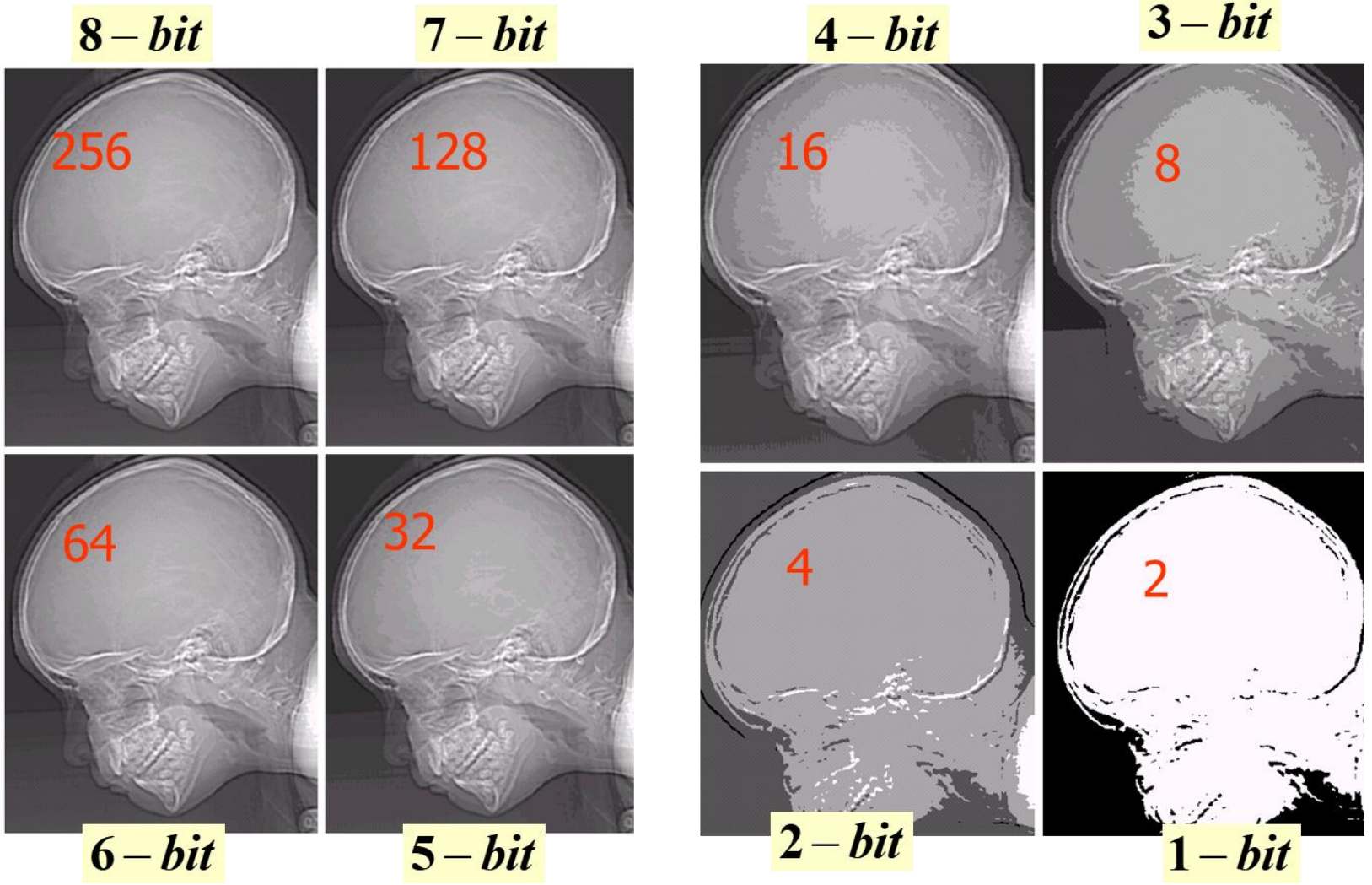


A value at each spatial location forms a two-dimensional array.

Quantization



Influence of Gray Scale Resolution



False Contour



Image Size and Storage

Must image dimensions be powers of two?

No. Powers of two are common, but other sizes are supported.
Efficient FFT lengths are not limited to powers of two.

How much storage for 512 × 512 pixels and 256 gray levels?

$k = \log_2 256 = 8$ bit/pixel

$b = M \times N \times k = 512 \times 512 \times 8$ bit

= 2,097,152 bit = 262,144 B = 256 KiB

Uncompressed, single-channel pixel data; headers and metadata excluded.

Sampling and Quantization

Non-uniform sampling of the image:

In areas where the gray level changes sharply, use delicate sampling, and in areas where the gray level is relatively smooth, use rough sampling.

Non-uniform quantization of the image:

Non-uniform quantization is based on the probability density function of the specific gray value distribution of an image, and quantization is performed according to the principle of minimum total quantization error.

The practical method is to set the quantization interval smaller for the frequently appearing gray value in the image and set the quantization interval larger for rarely appearing gray value.

Sampling versus Quantization

Fewer spatial samples

A coarser grid may lose fine structures or introduce aliasing.

Fewer gray levels

Coarser amplitude coding may produce banding and false contours.

Change one factor at a time when comparing results.
Keep bit depth fixed for sampling tests, and the grid fixed for quantization.

Interpolation Comparison

- Example:

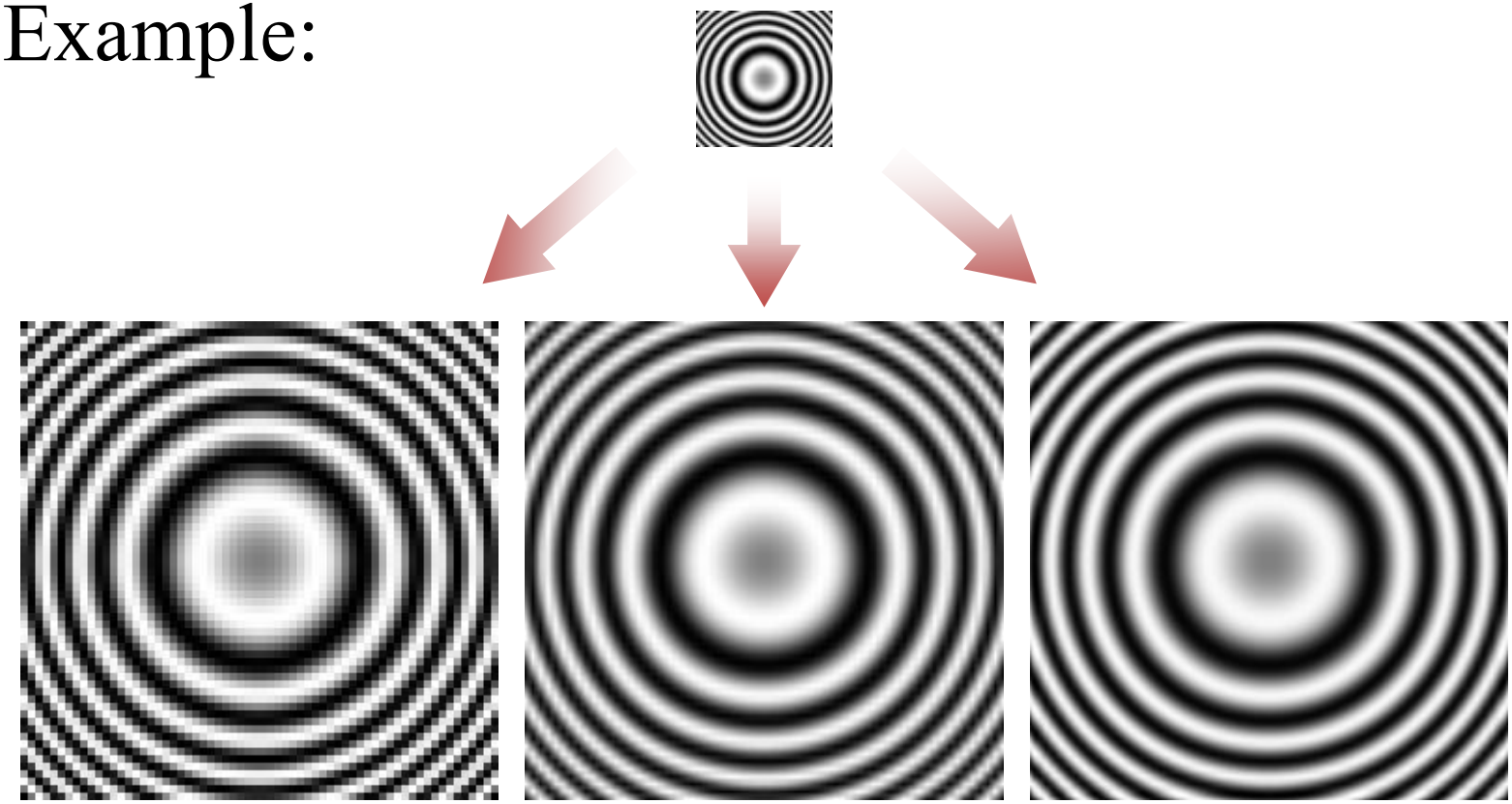


Image Scaling and Interpolation

Upsampling

Create a denser output grid and interpolate its pixel values.
More pixels do not automatically add acquired detail.

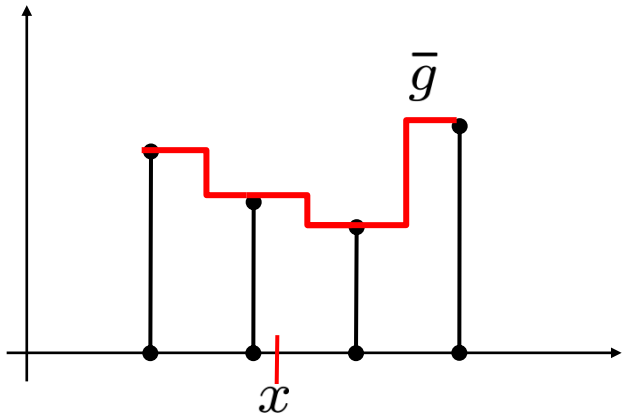
Downsampling

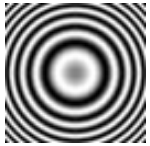
Apply a suitable low-pass filter before reducing the sampling rate.
Simply deleting rows and columns may cause aliasing.

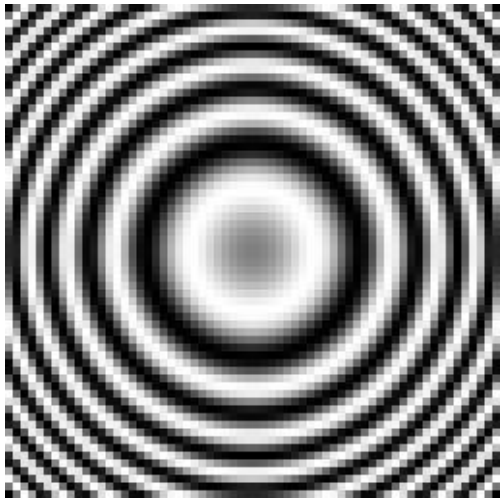
Medical use: linear interpolation is an option for gray-scale images.
Use label-preserving methods, such as nearest neighbor, for label maps.

Image Scaling and Interpolation

- changing the image size
- scaling needs evaluation of the image at non-integer positions → interpolation
- nearest neighbor interpolation:
  - approximating the grey level function with a step function
  - take the grey value of the nearest integer position
  - problem: blocking effect



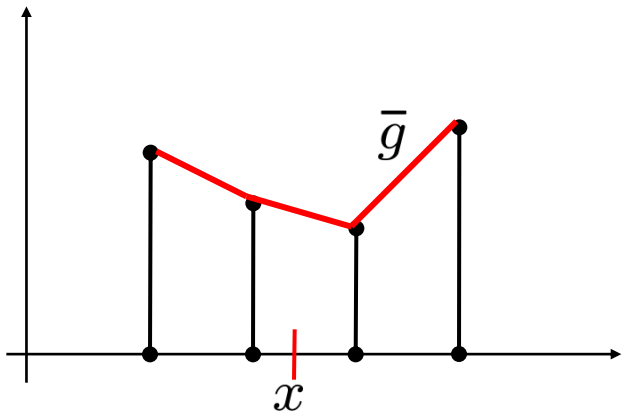




Interpolation

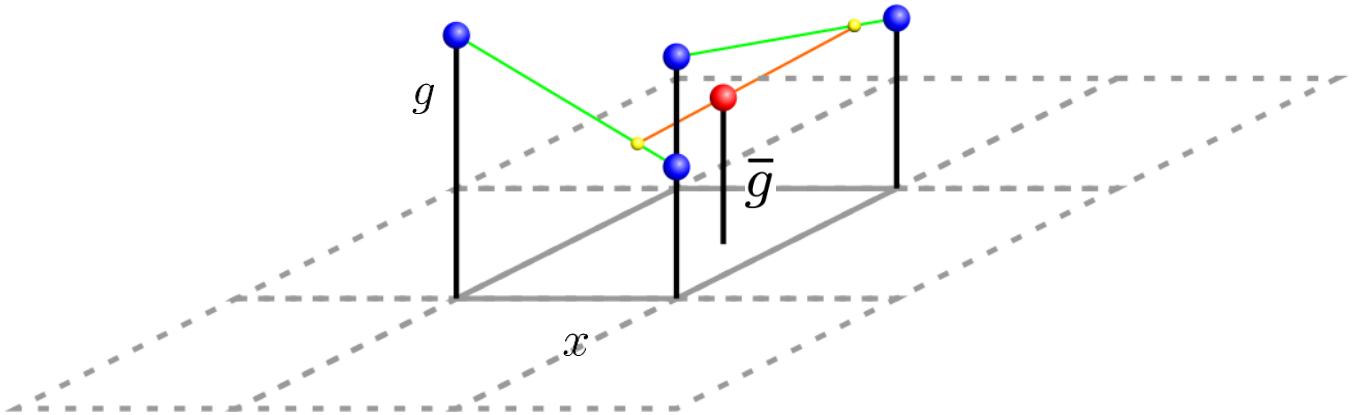
- Linear interpolation in 1D
  - fit linear function locally around x

$$\bar{g}(x) = g(\lfloor x \rfloor) + (x - \lfloor x \rfloor)(g(\lfloor x \rfloor + 1) - g(\lfloor x \rfloor))$$



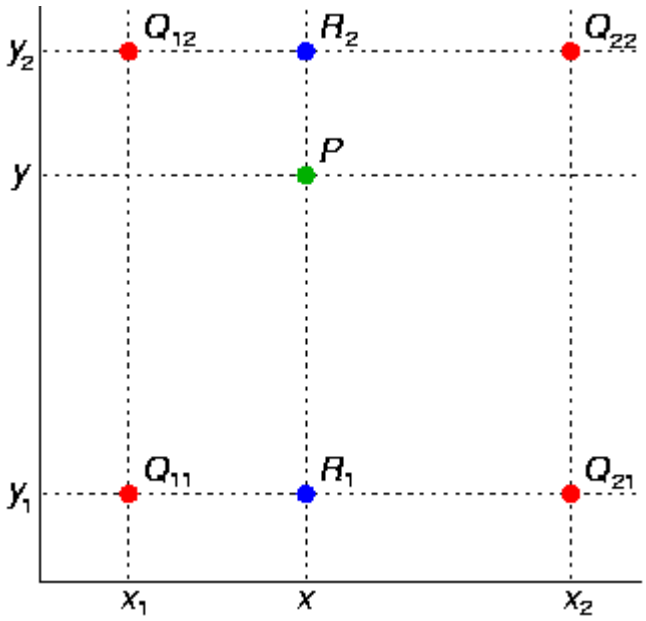
Bilinear Interpolation

- Extension of linear interpolation to 2D:



- Interpolate from 4 neighboring pixels
- Blocking is not obvious
- Degrading image details (smoothing effect)

Bilinear Interpolation



We first do linear interpolation in the *x*-direction

$$f(x, y_1) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21}),$$
$$f(x, y_2) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22}).$$

We proceed by interpolating in the y-direction to obtain the desired estimate.

Interpolation cont.

- cubic interpolation
  - fit cubic polynomial to the grey level
  - Solve

$$\bar{g}(x) = a \cdot (x - \lfloor x \rfloor)^3 + b \cdot (x - \lfloor x \rfloor)^2 + c \cdot (x - \lfloor x \rfloor) + d$$

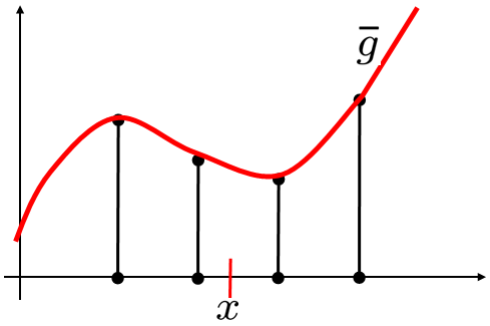
yields:

$$a = -\frac{1}{6}g(\lfloor x \rfloor - 1) + \frac{1}{2}g(\lfloor x \rfloor) - \frac{1}{2}g(\lfloor x \rfloor + 1) + \frac{1}{6}g(\lfloor x \rfloor + 2)$$

$$b = \frac{1}{2}g(\lfloor x \rfloor - 1) - g(\lfloor x \rfloor) + \frac{1}{2}g(\lfloor x \rfloor + 1)$$

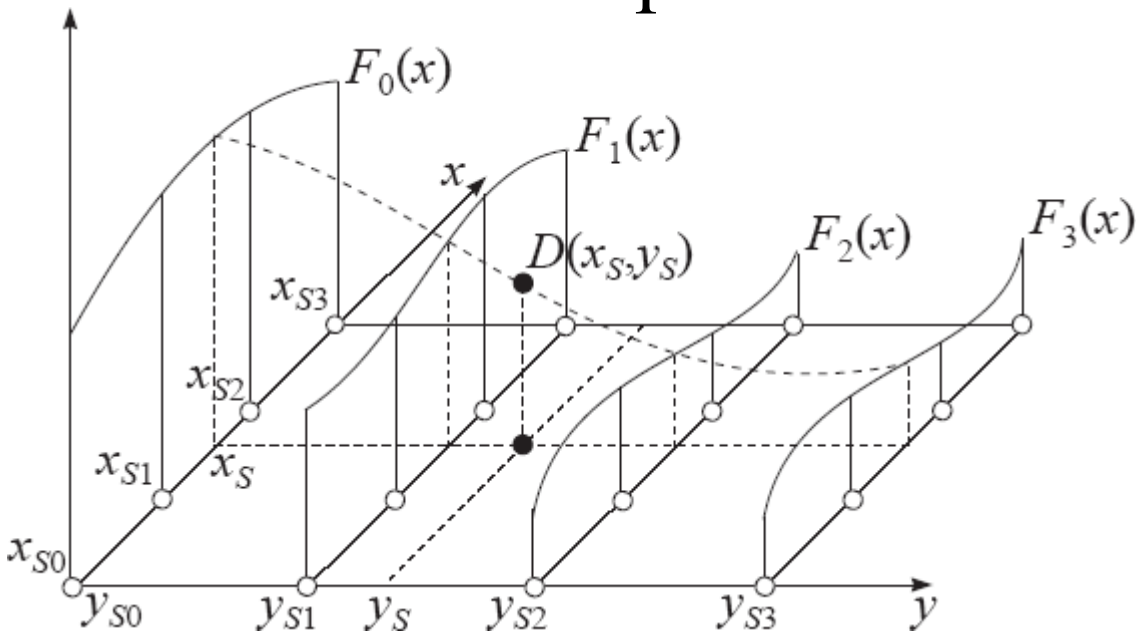
$$c = -\frac{1}{3}g(\lfloor x \rfloor - 1) - \frac{1}{2}g(\lfloor x \rfloor) + g(\lfloor x \rfloor + 1) - \frac{1}{6}g(\lfloor x \rfloor + 2)$$

$$d = g(\lfloor x \rfloor)$$



Interpolation cont.

- extension of cubic interpolation to 2D:



- interpolation from 16 neighboring pixels

Summary and Self-check

Sampling sets locations; quantization represents amplitudes.

Consider anti-aliasing before downsampling. Interpolation refines the output grid without automatically adding acquired detail.

Distinguish intensity data, display mappings, and segmentation labels.

1. Do more pixels necessarily mean more true detail?
2. What artifacts can appear when bit depth is reduced?
3. Why can ordinary linear interpolation be unsuitable for labels?



Pixel Relationships and Image Operations

How do pixels form an object?

Adjacency and connectivity · Distance · Image operations

Applications: microscopy, cell masks and medical imaging

How Many Objects Are in This Mask?

Opening question | Simulated nuclear mask

1	1	0	0	0
1	1	0	0	0
0	0	1	1	0
0	0	1	1	0
0	0	0	0	0

1 = foreground
0 = background

The two blocks touch only at a corner.

How many objects would you count?

Define connectivity before counting objects.

Learning Objectives (section2)

By the end of this lesson, you should be able to:

Determine pixel adjacency and connectivity.

Calculate three distances and convert pixels to physical units.

Select operations for denoising, subtraction, shading and masking.

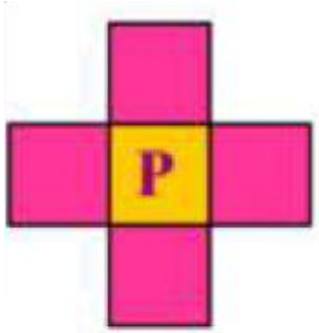
Use each definition to solve a judgment or calculation problem.

Basic Relationship between Pixels

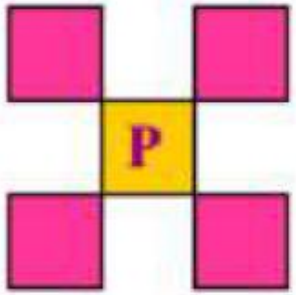
- Basic relationship between pixels
  - Adjacent: 4-Neighbor, D-Neighbor, 8-Neighbor
  - Distance metric
  - Adjacency, connectivity
  - Area, boundary

Pixel Neighborhoods(邻域)

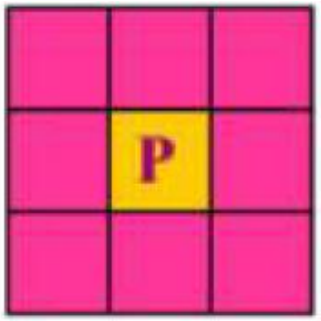
The center pixel p=(x,y) is not its own neighbor



4-neighbors



Diagonal neighbors



8-neighbors

$N_4(p)$: four horizontal and vertical neighbors

$N_D(p)$: four diagonal neighbors

$N_8(p) = N_4(p) \cup N_D(p)$ — the union of both sets

Adjacency（邻接）: Position and Pixel Values

First define the set of allowed intensity values V

Required: f(p) ∈ V and f(q) ∈ V

Rule	Spatial condition
4-adjacency	$q \in N_4(p)$
8-adjacency	$q \in N_8(p)$

Binary: V={1}; intensity threshold: V={128,…,255}

4-adjacency implies 8-adjacency; the converse need not hold.

m-Adjacency: The Diagonal Rule

Mixed adjacency | Apply the shared-neighbor test only to diagonal neighbors

Required: $f(p) \in V$ and $f(q) \in V$

Case	Spatial condition
4-neighbor	$q \in N_4(p)$
Diagonal neighbor	$q \in N_D(p)$, and $N_4(p) \cap N_4(q)$ contains no pixel with a value in V

Either spatial condition is sufficient, provided $f(p), f(q) \in V$.

Check shared 4-neighbors only for the diagonal case.

m-Adjacency: Shared Neighbors

Simulated example | V={1}; coordinates are (row, column)

1 (p)	1
0	1 (q)

A shared neighbor is 1.
Diagonal step blocked.

1 (p)	0
0	1 (q)

Shared neighbors are 0.
Diagonal step allowed.

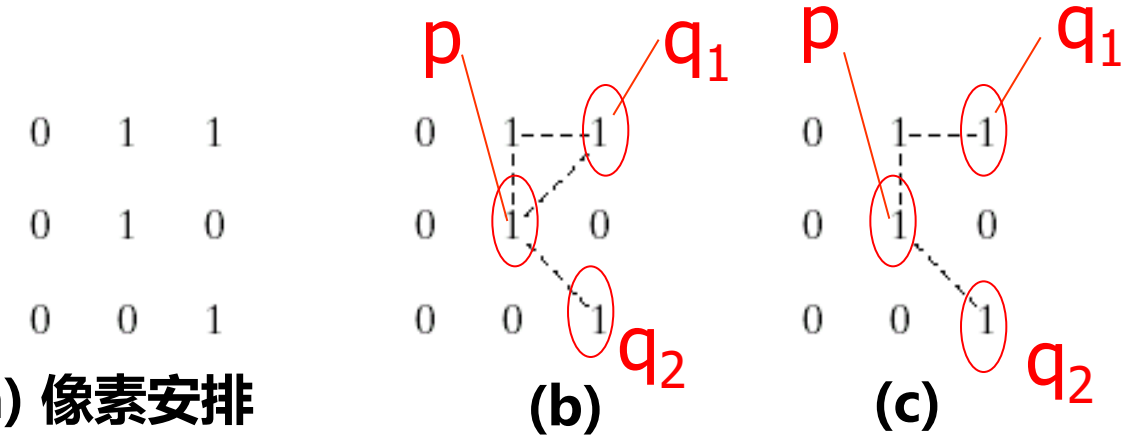
On the left, p and q remain connected via the third foreground pixel.

Ambiguities

✓只定义8邻接，则q_2和q_1之间的通路有两条（二义）见 （b）；

✓定义了m邻接，则q_2和q_1之间的通路就只有一条（m通路）见（c）

V={1}



a b c (a) 像素安排 (b) (c)

(b) 中心像素p的8邻接像素：q_1, q_2

(c) 中心像素p的m邻接像素：q_2

q_1和p:8邻接，非m邻接　q_2和p:8邻接，又m邻接

不满足条件：N4(p)∩N4(q1)没有V值的像素

Paths (通路) and Connectivity (连接)

Path and connectivity

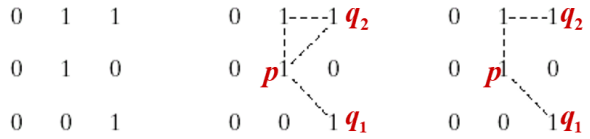
Path: $p_0, p_1, \ldots, p_n$; consecutive pixels satisfy the adjacency rule.

Path length = n steps; the sequence has n+1 positions.

p and q are connected if a path within S joins them.

A path is closed when its first and last pixels coincide.

Pixels can be connected without being directly adjacent.



a b c

(b)图中，q_1和q_2 之间存在2条通路；
(c)图中，q_1和q_2 之间只有1条通路；（m通路）

Connected Components

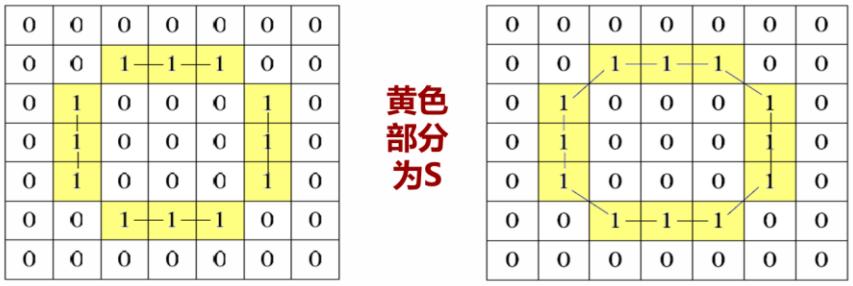
Connected component

Choose a foreground set S and an adjacency rule.

Group pixels that can reach each other along paths within S.

Each maximal connected group is a connected component.

A region can be defined as a connected set of pixels.



（a）4连通 连通分量:4 其中v={1} 连通分量:1（b）8连通

Class Exercise: Count the Objects Again

Work alone for 1 minute, then discuss for 1 minute

1	1	0	0	0
1	1	0	0	0
0	0	1	1	0
0	0	1	1	0
0	0	0	0	0

V={1}

① How many 4-connected components?

② How many 8-connected components?

③ Does each component represent one nucleus?

Identify the pair of pixels that changes the count.

Answer: Connectivity Changes the Count

4-connectivity: 2 components; 8-connectivity: 1 component

Connectivity	Components	Reason
4-connected	2	No horizontal or vertical connection
8-connected	1	A diagonal joins the two blocks

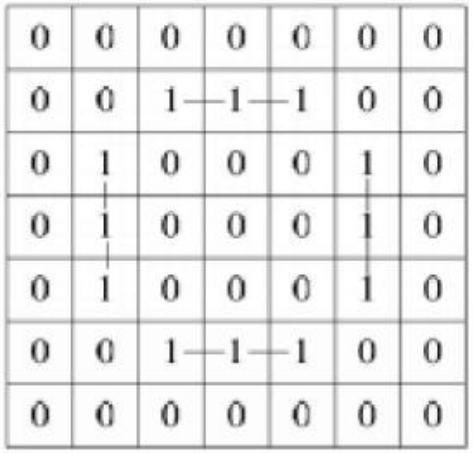
Key pixels: (row 2, column 2) and (row 3, column 3).

Real nuclei may merge or split in a segmented mask.
Component counts do not automatically equal nuclear counts.

Report both the mask-generation method and the connectivity rule.

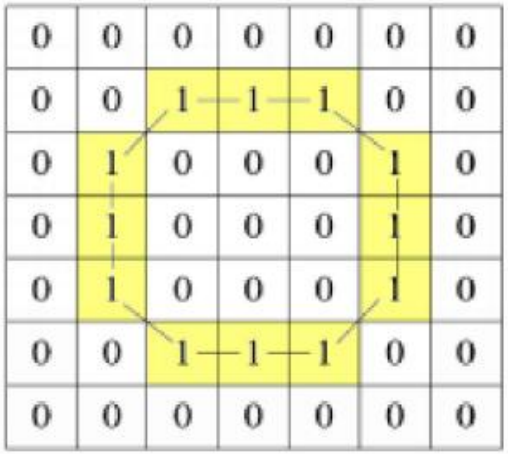
Region （区域）

区域：R是图像中的像素子集。如果R是连通集，则称R为一个区域（黄色部分）。



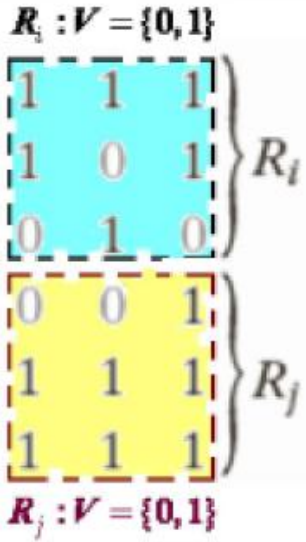
（a）4连通

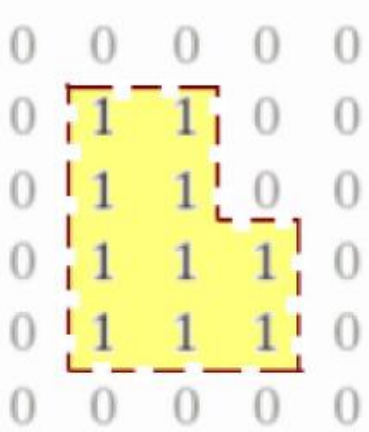
其中v={1}



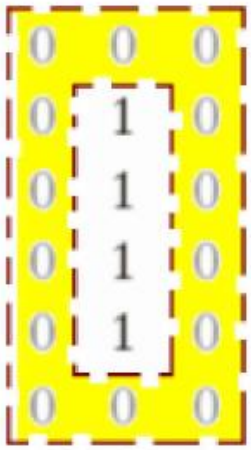
（b）8连通

邻接区域：两个区域。如果联合（并）为一个连通集，则称这两个区域为邻接区域。如图R_i和R_j





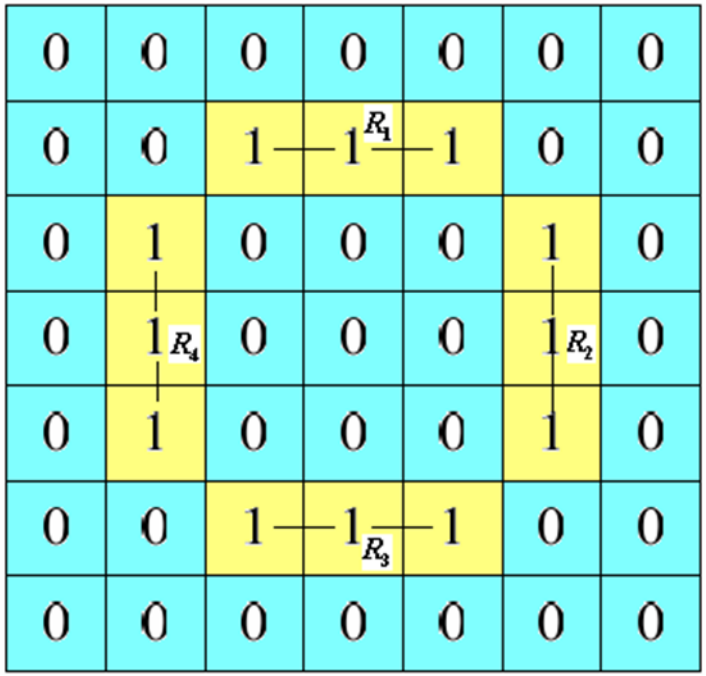
$V = \{1\}$



$V = \{0\}$

Boundary(边界)

假设一副图像S中有K个不连接区域，且它们都不接触图像边界。即：$R_k, k = 1, 2, \cdots, K$



前景：$R_u = R_1 \cup R_2 \cup R_3 \cup \cdots \cup R_K$

背景：$(R_u)^c = S - R_u$

内边界：

一个区域的边缘或轮廓线叫做边界。

即：该区域中和其背景相邻接的点的集合

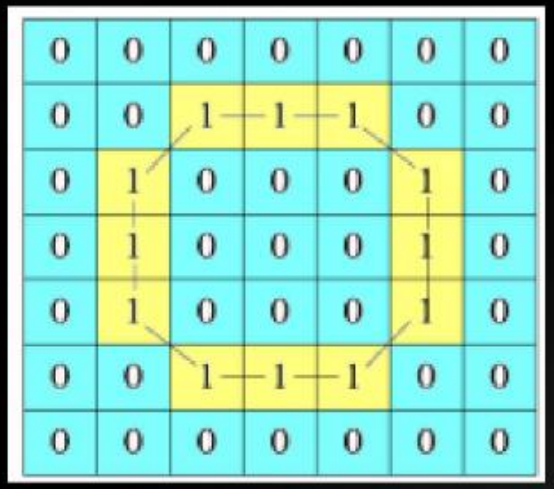
外边界：对应于背景边界。

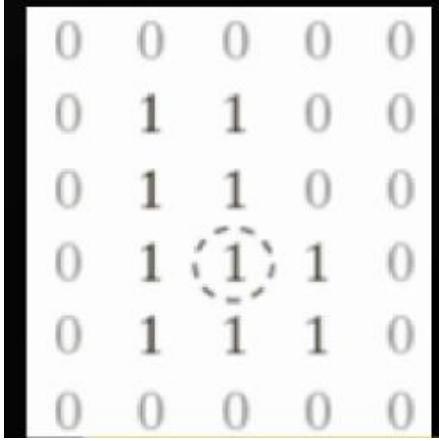
Boundary(边界)

本例用 8 邻域判断区域与背景的接触。
边界定义须先明确所采用的邻域规则。

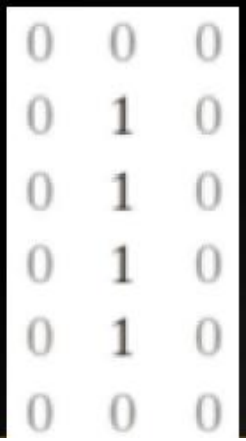
前景（黄色区域）：V={1}，内边界就是它自身；

背景（兰色区域）：V={0}





左图中，被圈出的点如果在区域及背景间使用4连通，就不是1值区域边界的成员



左图中，1值区域的内边界就是区域自身（不是闭合通路），而外边界是一个围绕该区域的闭合通路

Distance Measures

对于象素点 $p(x,y)$, $q(s,t)$, $z(v,w)$, 如果

(a) $D(p,q) \geq 0$ $\left[p=q\text{时}, D(p,q)=0\right]$ 正定性

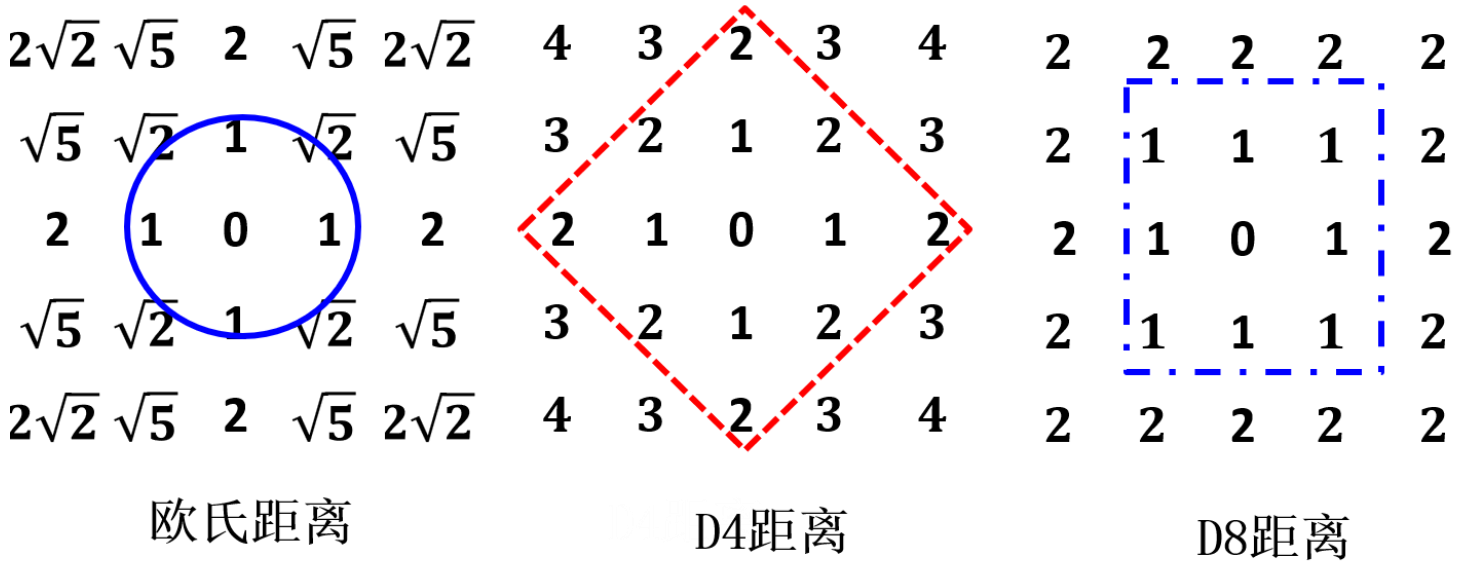
(b) $D(p,q)=D(q,p)$ 对称性

(c) $D(p,z) \leq D(p,q)+D(q,z)$ 距离三角不等式

则D是距离的度量函数.

Distance	Definition
Euclidean D_e	$\sqrt{[(x-s)^2 + (y-t)^2]}$
City-block D_4	$\|x-s\| + \|y-t\|$
Chessboard D_8	$\max(\|x-s\|, \|y-t\|)$

Distance Measures



欧氏距离　　D4距离　　D8距离

On a unit square grid: $D_8(p,q) \le D_e(p,q) \le D_4(p,q)$

Distance	Definition
Euclidean D_e	$\sqrt{[(x-s)^2 + (y-t)^2]}$
City-block D_4	$\lvert x-s\rvert + \lvert y-t\rvert$
Chessboard D_8	$\max(\lvert x-s\rvert, \lvert y-t\rvert)$

Distance Example: One Pair, Three Metrics

Simulated example | p=(1,1), q=(4,5)

Horizontal separation: 3; vertical separation: 4.

Metric	Calculation	Answer
D_e	$\sqrt{(3^2+4^2)}$	5
D_4	3+4	7
D_8	max(3,4)	4

Paths restricted to a mask must also account for obstacles.

Microscopy: Pixels to Physical Distance

Simulated example | Sample-plane spacing: $\Delta x = 0.2\ \mu m$, $\Delta y = 0.4\ \mu m$

Horizontal displacement: $3 \times 0.2 = 0.6\ \mu m$

Vertical displacement: $4 \times 0.4 = 1.6\ \mu m$

Physical distance = $\sqrt{(0.6^2 + 1.6^2)} \approx 1.71\ \mu m$

Read or calibrate the pixel spacing before reporting µm.

Mathematical Tools

- Array versus Matrix Operations
- Linear versus Nonlinear Operations
- Arithmetic Operations
- Set and Logical Operations

Array versus Matrix Operations

- An array operation involving one or more images is carried out on a **pixel by pixel** basis

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

- A matrix operation is carried out using matrix theory

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{12}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Linear versus Nonlinear Operations

- H is Linear Operator if

$$H(a_i f_i + a_j f_j) = a_i H(f_i) + a_j H(f_j)$$

Sum operator $\sum$:

$$\sum_i \left[a f_i(x,y) + b g_i(x,y) \right] = a \sum_i f_i(x,y) + b \sum_i g_i(x,y)$$

Max operator max:

$$f_1(x,y) = \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} \quad f_2(x,y) = \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix}$$

$$\max\{f_1 - f_2\} = \max\left\{\begin{bmatrix} -6 & -3 \\ -2 & -4 \end{bmatrix}\right\} = -2 \neq \max\{f_1\} + (-1)\max\{f_2\} = 3 + (-1)\cdot 7 = -4$$

Arithmetic Operations

- Arithmetic operations are carried out between corresponding pixel pairs.

$$s(x,y) = f(x,y) + g(x,y)$$
$$d(x,y) = f(x,y) - g(x,y)$$
$$p(x,y) = f(x,y) \times g(x,y)$$
$$v(x,y) = f(x,y) \div g(x,y)$$

- Arithmetic operations are performed between corresponding pixel pairs.
- Images are the same in size.

Noise Reduction by Addition

$$g(x,y) = f(x,y) + \eta(x,y)$$

Corrupted image **Ideal image** **Noise**

Assume $\eta(x,y)$ follows the normal Gaussian distribution: $\eta(x,y) \sim N\left(0, \sigma^2_{\eta}(x,y)\right)$

By adding a set of noisy images $g_i(x,y)$: $\bar{g}(x,y) = \frac{1}{k}\sum_{i=1}^{k} g_i(x,y)$

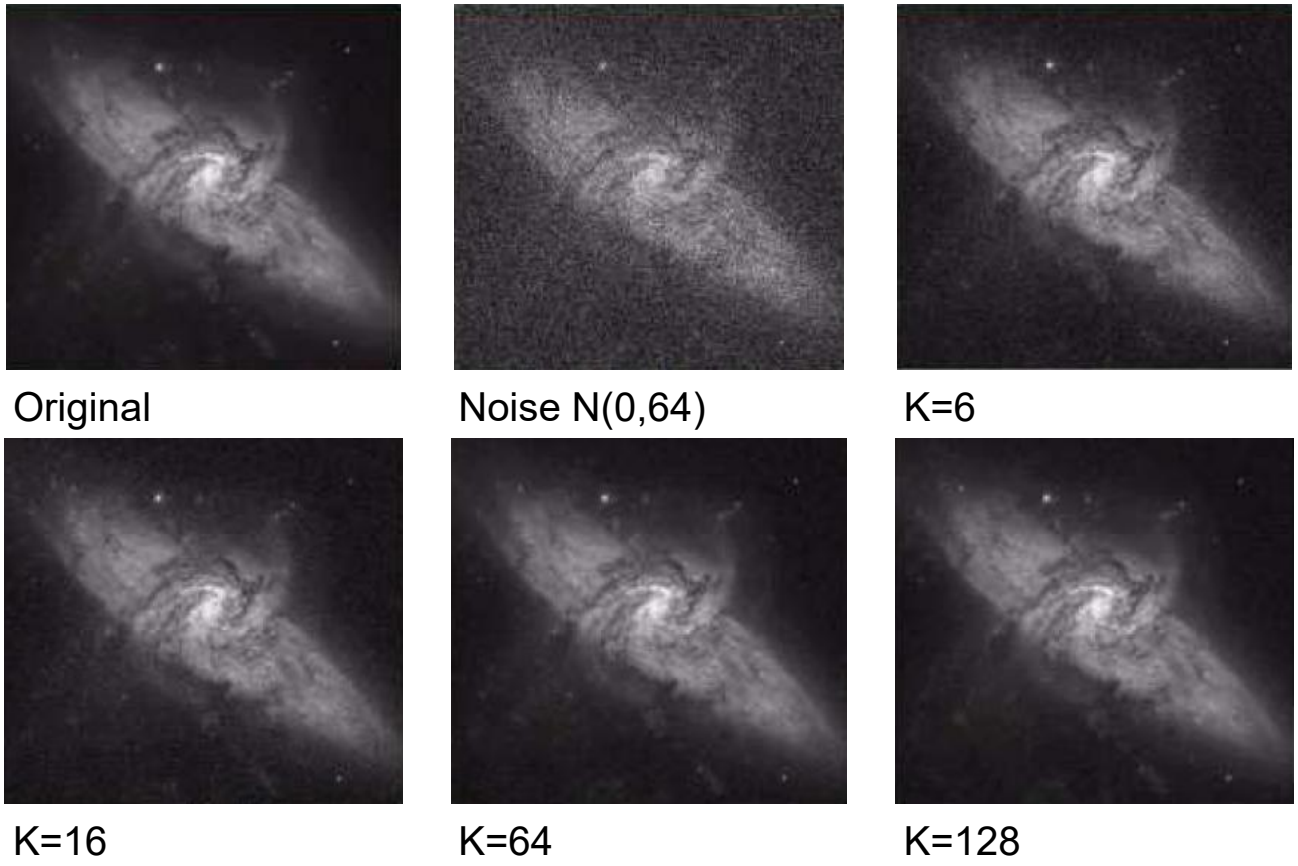
Expected value of $\bar{g}(x,y)$: $E\{\bar{g}(x,y)\} = f(x,y)$

The standard deviation at any point in the average image is: $\sigma_{\bar{g}(x,y)} = \frac{1}{\sqrt{k}}\sigma_{\eta}(x,y)$

For aligned images of a stable scene, assume zero-mean noise with equal variance and no correlation across frames.
As K increases, the variability (as measured by the standard deviation) of the pixel values at each location (x, y) decreases. This means that $\bar{g}(x,y)$ approaches $f(x,y)$

What Changes as More Frames Are Averaged?

Additive-noise demonstration from the source lecture



Discuss: can the same averaging approach handle moving live cells?

Image Addition

- For two images $f(x,y)$ and $h(x,y)$, find the mean value:

$$g(x,y)=\frac{1}{2}f(x,y)+\frac{1}{2}h(x,y)$$ **Double exposure effect**

$$g(x,y)=\alpha\cdot f(x,y)+\beta\cdot h(x,y) \quad \text{其中：} \alpha+\beta=1$$

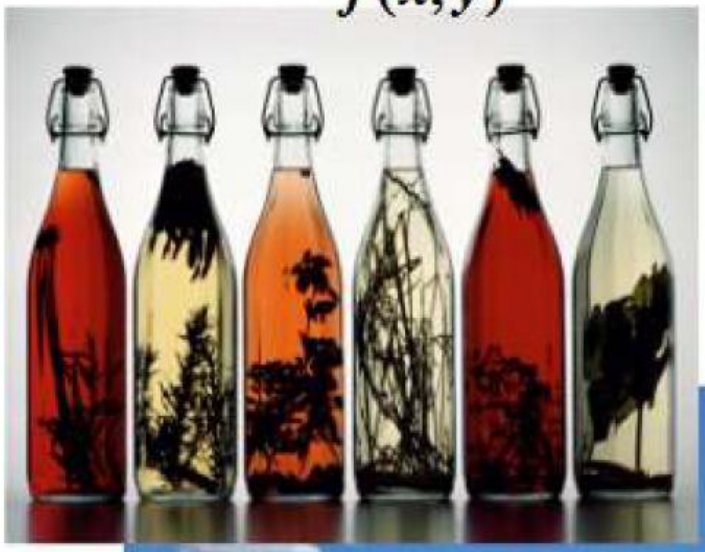
Various image synthesis effects, and can also be used for the connection of two pictures

Image Addition

$$g(x,y)=\alpha\cdot f(x,y)+\beta\cdot h(x,y) \quad \text{其中：} \quad \alpha+\beta=1$$

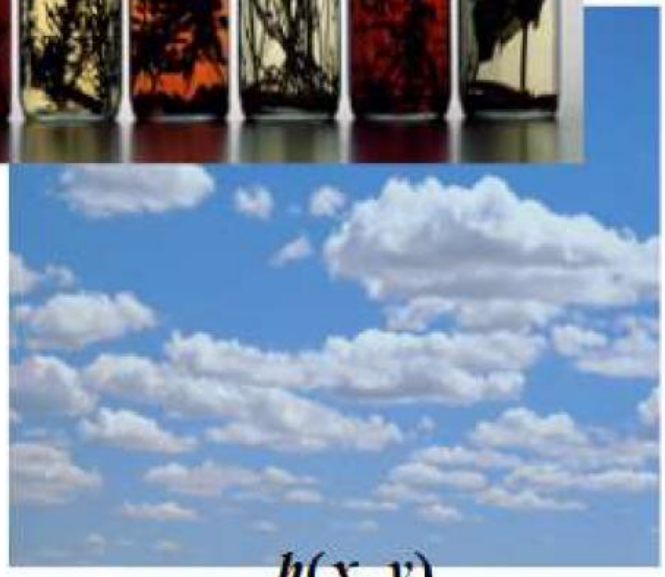
加法生成图象
的叠加效果

$f(x,y)$



$g(x,y)$





$h(x,y)$

Image Subtraction: Keep the Sign

Subtraction | $g(x,y)=f(x,y)-h(x,y)$

Pixel	f	h	g=f−h
A	100	80	20
B	80	100	−20
C	90	90	0

Compute in signed or floating-point format; set display separately.

Negative differences carry information and should be preserved.

Difference Images: Values Versus Display

Display mapping | For visualization only

When gmax > gmin:

$$gdisplay = 255 \times (g - gmin) / (gmax - gmin)$$

Raw difference	**−20**	**0**	**20**
Display value	0	127.5	255

Handle constant images separately; keep raw differences for analysis.

Image Subtraction

- Remove unnecessary superimposed patterns

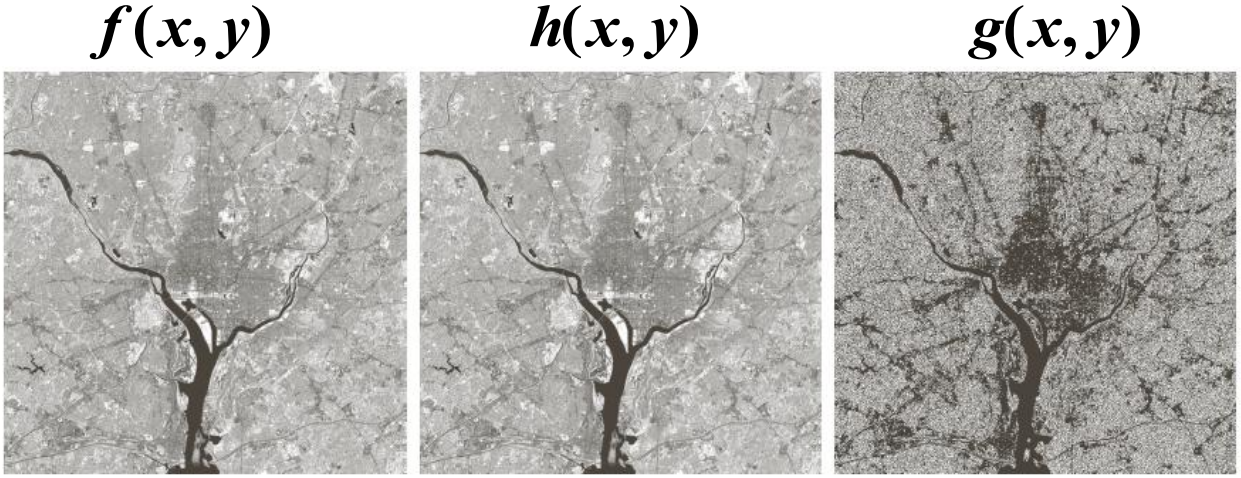
Suppose: background image b(x,y), foreground and background mixed image f(x,y)

$$g(x,y) = f(x,y) - b(x,y)$$

g(x,y) is the image with the background removed.

Image Subtraction

- Detect the difference between two images;



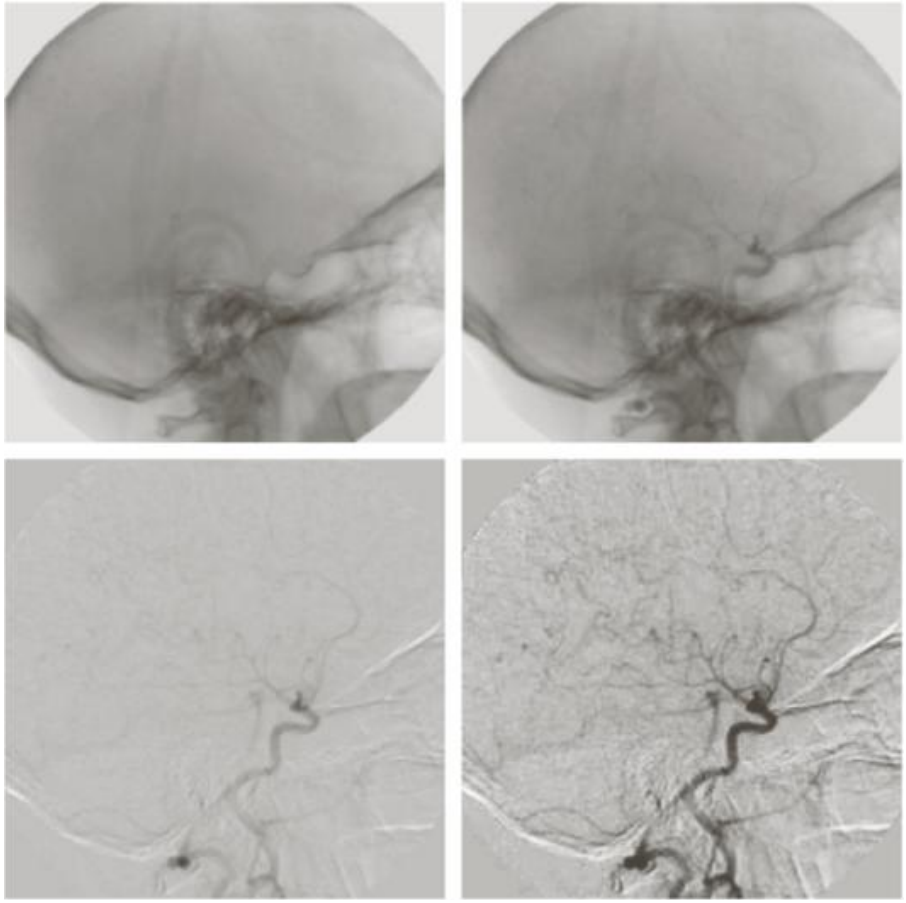
a b c

FIGURE 2.27 (a) Infrared image of the Washington, D.C. area. (b) Image obtained by setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range $[0, 255]$ for clarity.

- (b):Set the lowest-order bit position of each pixel in (a) to 0; (lower-order bits contain more minute details in the image)
- (c) :The black area indicates that there is no difference between the two images. Negative gray values may appear, so calibration is required.

Medical Example: Subtraction Angiography

Digital subtraction angiography



Top row:
Reference | Post-contrast

Bottom row:
Difference | Enhanced difference

Question:
How would motion affect the difference image?

Check alignment before subtraction.

Image Subtraction

- Detect the difference between two images;

Suppose: the image at time 1 is T1(x,y),the image at time 2 is T2(x,y).

$$g(x,y) = T2(x,y) - T1(x,y)$$

$T2(x,y)$



$T1(x,y)$



$g(x,y)$



Neighbor Differences: Estimate the Gradient

Unit pixel spacing; finite differences approximate derivatives

$G_x = f(x+1,y) - f(x,y)$

$G_y = f(x,y+1) - f(x,y)$

Gradient magnitude $\approx \sqrt{(G_x^2 + G_y^2)}$

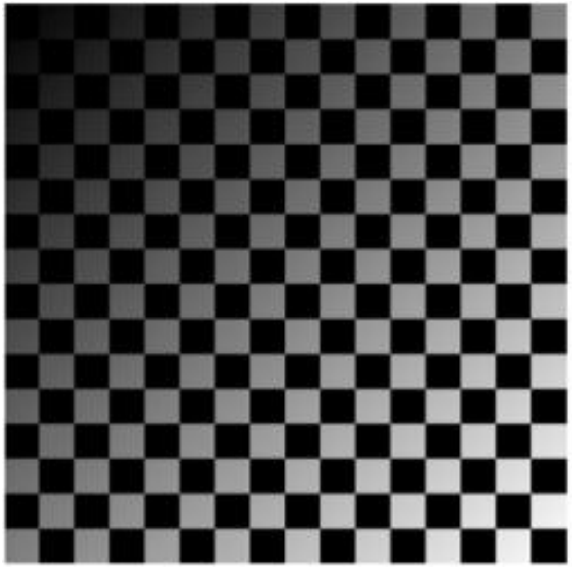
Simplified approximation: $\max(|G_x|, |G_y|)$

Example: $G_x=-3$, $G_y=-4$ → magnitude 5; max approximation 4.

Image Multiplication(and Division)

Simplified model: $I(x,y)=F(x,y)\cdot L(x,y)$, with $L>0$

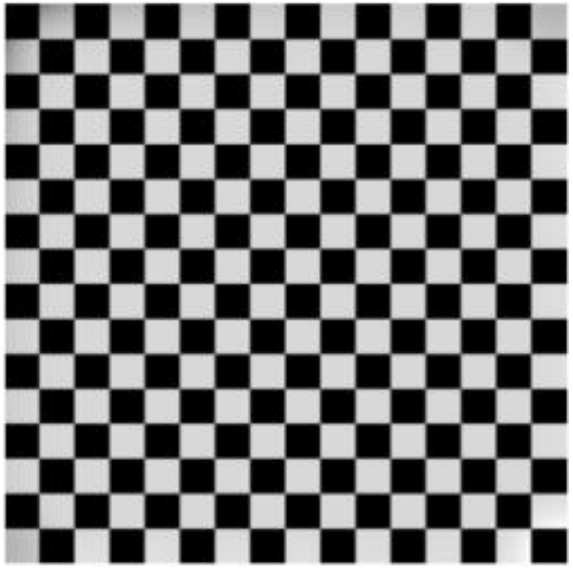
Observed image I



Illumination L



Corrected image F



$$F = I / L = I \times (1/L)$$

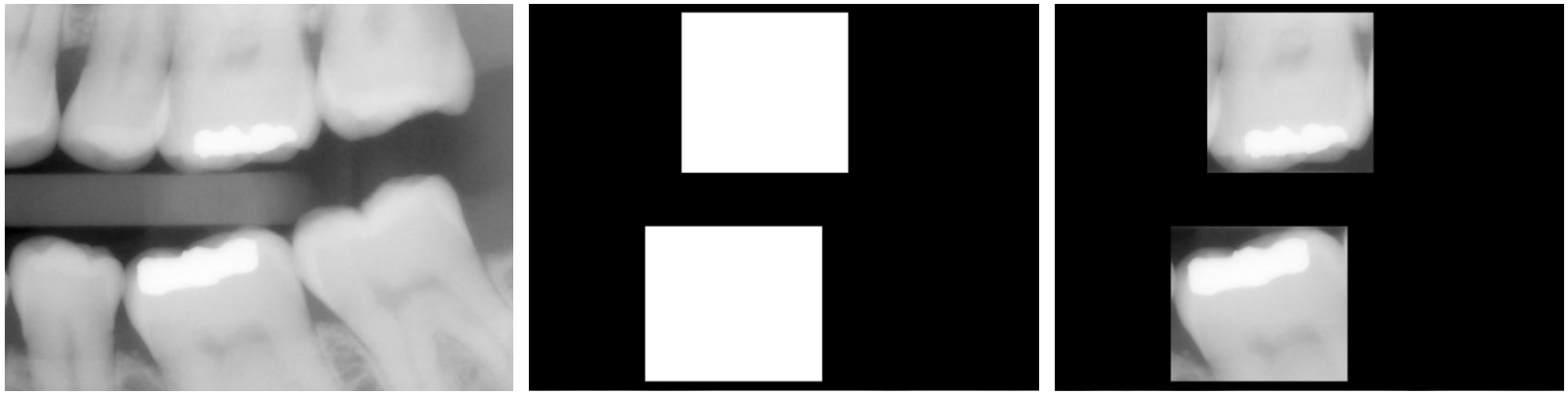
Subtract additive background; divide by multiplicative illumination.

Image Multiplication(and Division)

Multiplication: $C(x,y)=A(x,y)\times B(x,y)$

Applications:

- Masking



a b c

FIGURE 2.34 (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

Multiply by 1 to keep or 0 to suppress; the image size stays the same.

Arithmetic Operations (Combined)

Q: Given an image f, how to guarantees that the full range of an arithmetic operation between images is "captured" into a fixed number of bits?

- 1. Creates an image whose minimum value is 0:

$$f_m = f - \min\{f\}$$

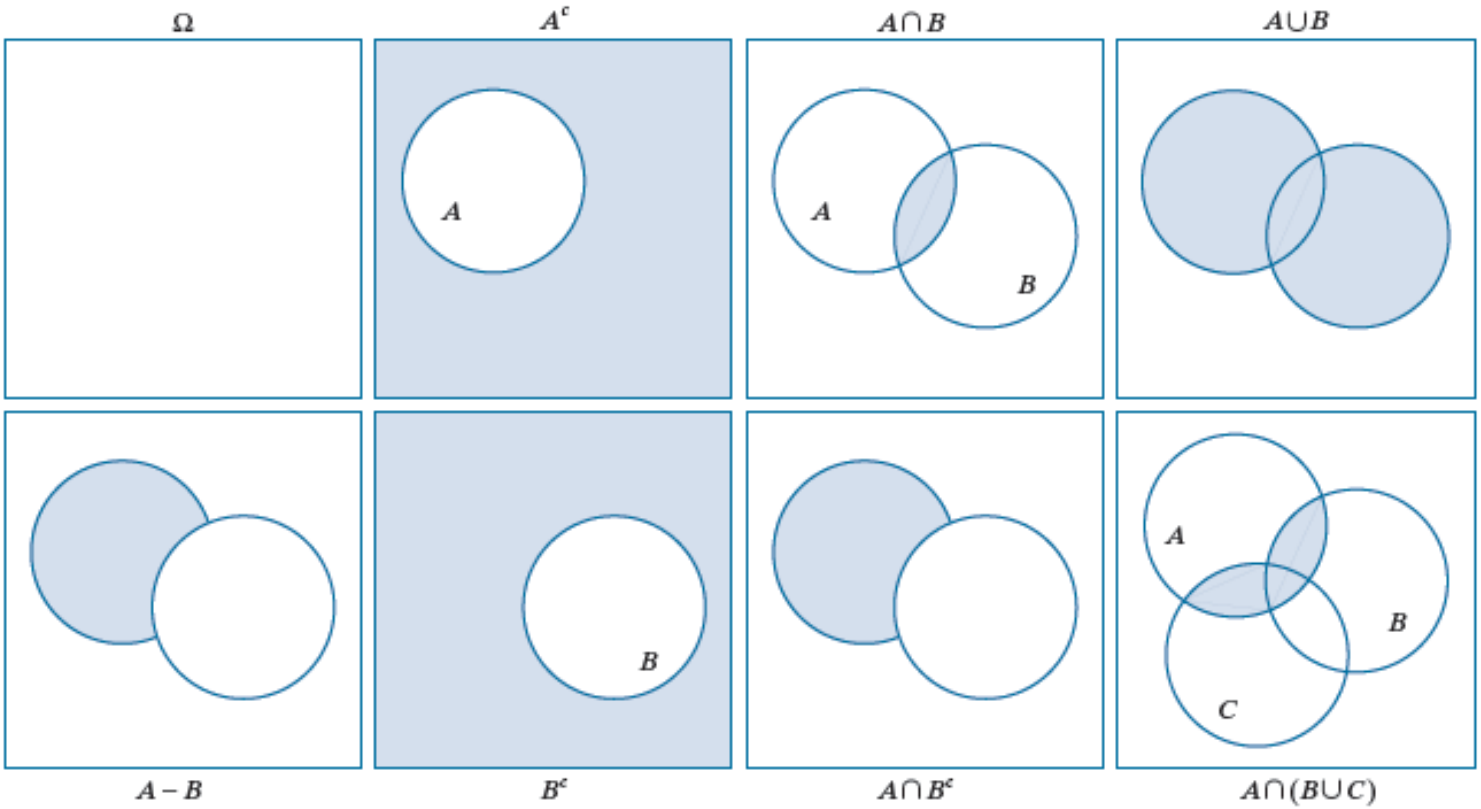
- 2. Creates a scaled image, f_s, whose values are in the range [0 ,K]:

$$f_s = K\left[f_m / \max\{f_m\}\right]$$

If max(fm)=0, the image is constant: use a uniform display.
Keep the original values for quantitative analysis.

Set Operations

- Basic set operations



a b c d
e f g h

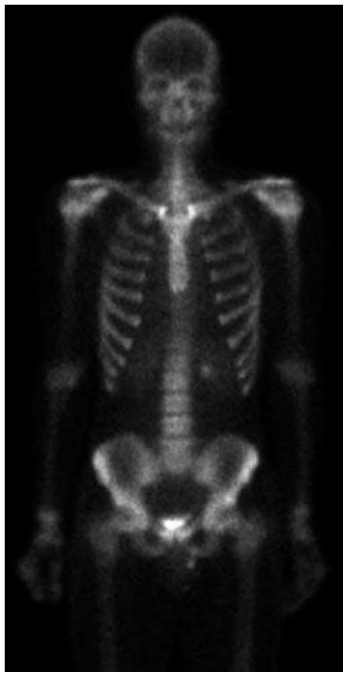
FIGURE 2.35 Venn diagrams corresponding to some of the set operations in Table 2.1. The results of the operations, such as A^c, are shown shaded. Figures (e) and (g) are the same, proving via Venn diagrams that $A - B = A \cap B^c$ [see Eq. (2-40)].

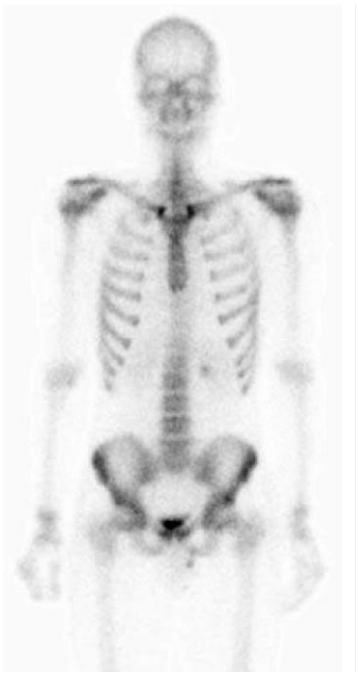
Set Operations

- Set operations in grey-scale images

  - Complement:

    $A^c = \{(x, y, K - z) | (x, y, z) \in A\}$,
    where x and y are spatial coordinates and z denotes intensity

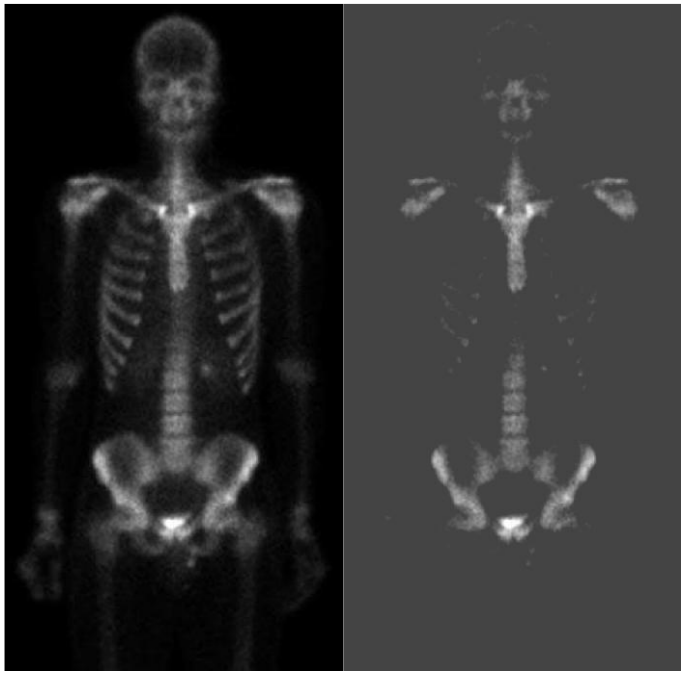




Set Operations

- Set operations in grey-scale images
  - Union:

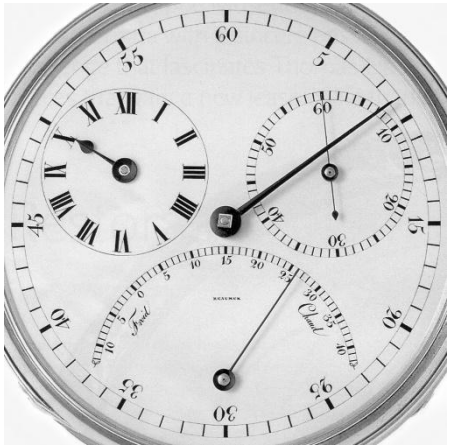
$$A \cup B = \left\{ \max_{z}(a, b) \,\middle|\, a \in A, b \in B \right\}$$

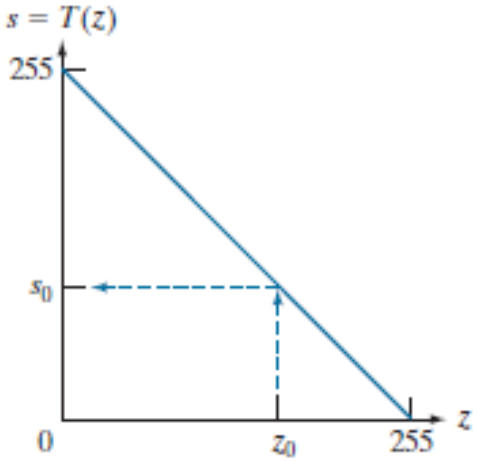


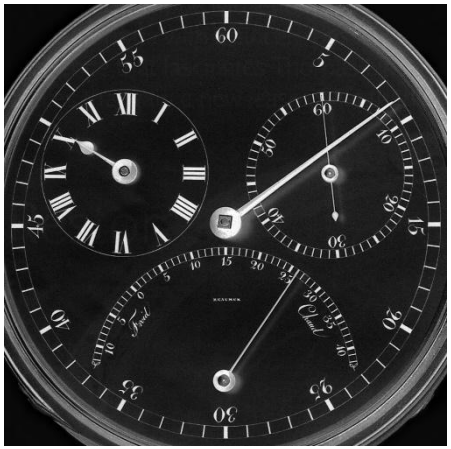
Spatial Operations

- Single-pixel operation

$$s = T(z)$$



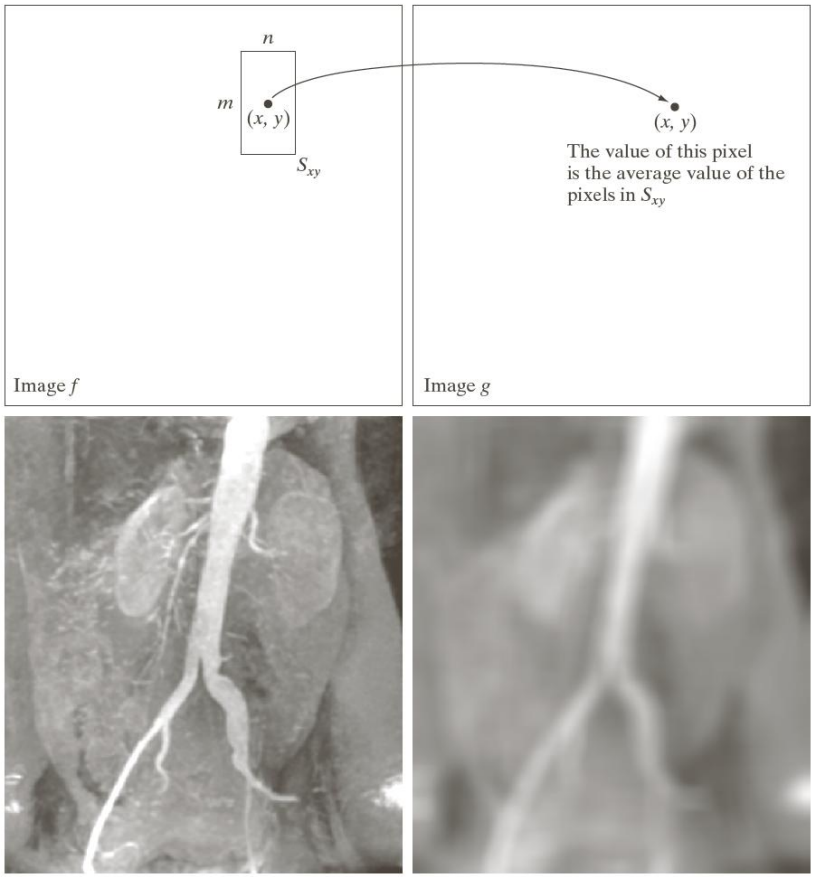




Spatial Operations

Neighborhood operations

Local averaging: $$g(x,y) = \frac{1}{mn} \sum_{(r,c) \in S_{xy}} f(r,c)$$



a b
c d

FIGURE 2.39 Local averaging using neighborhood processing. The procedure is illustrated in (a) and (b) for a rectangular neighborhood. (c) An aortic angiogram (see Section 1.3). (d) The result of using Eq. (2-43) with $m = n = 41$. The images are of size 790×686 pixels. (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

Spatial Operations

Geometric Spatial Transformations

TABLE 2.3 Affine transformations based on Eq. (2-45).

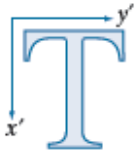
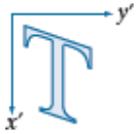
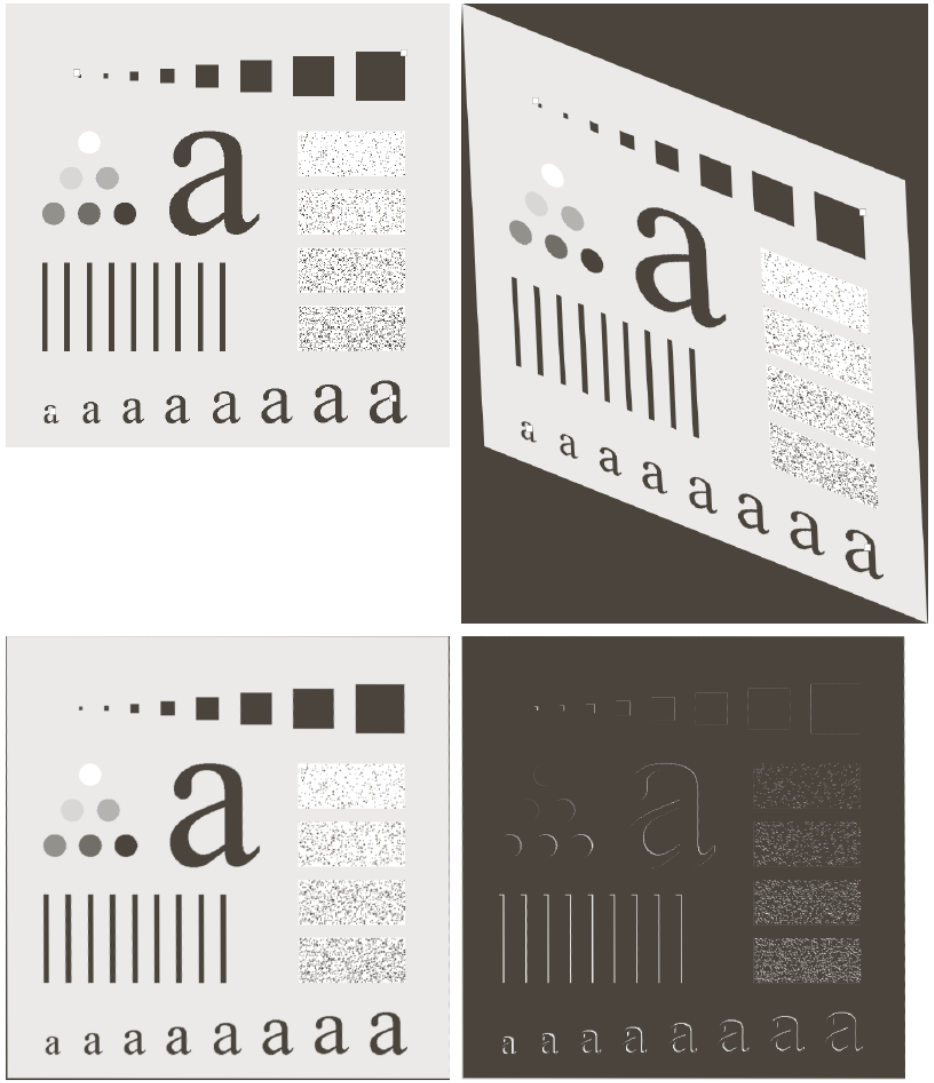
Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
Scaling/Reflection (For reflection, set one scaling factor to −1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
Rotation (about the origin)	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x\cos\theta - y\sin\theta$ $y' = x\sin\theta + y\cos\theta$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$	

Image Registration



a b
c d

FIGURE 2.42
Image registration. (a) Reference image. (b) Input (geometrically distorted image). Corresponding tie points are shown as small white squares near the corners. (c) Registered (output) image (note the errors in the border). (d) Difference between (a) and (c), showing more registration errors.

Vector and Matrix Operations

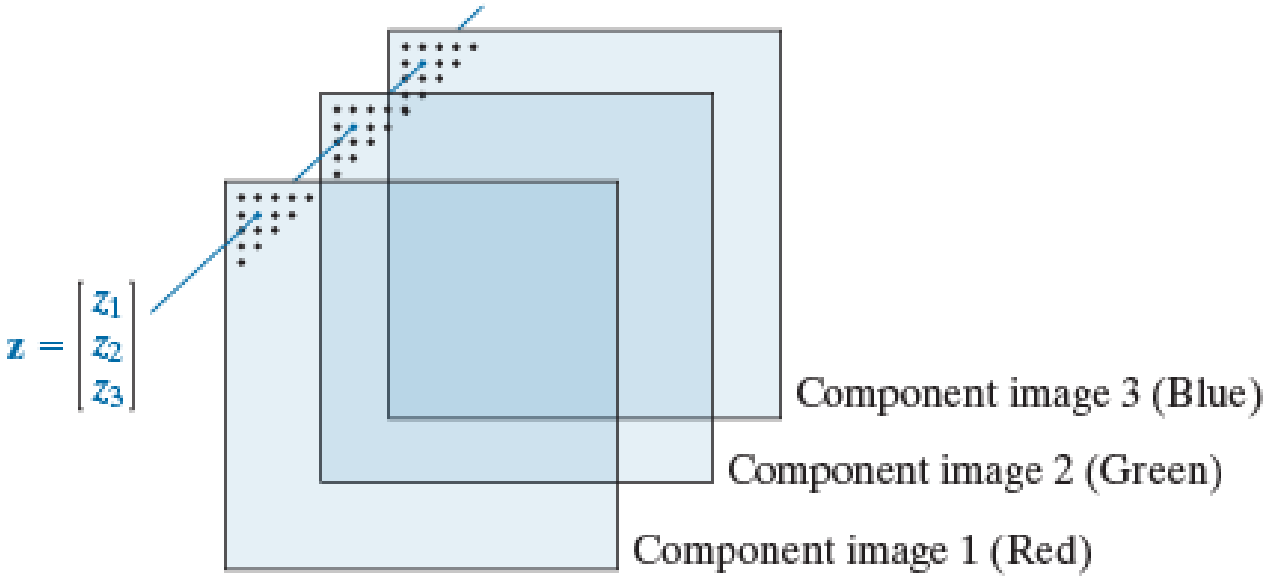


FIGURE 2.43 Forming a vector from corresponding pixel values in three RGB component images.

数字图像基础

2.1 视觉感知要素

2.2 光和电磁波谱

2.3 图像感知和获取

2.4 图像采样和量化

2.5 像素间关系和基本操作

2.6 图像处理的数学工具介绍